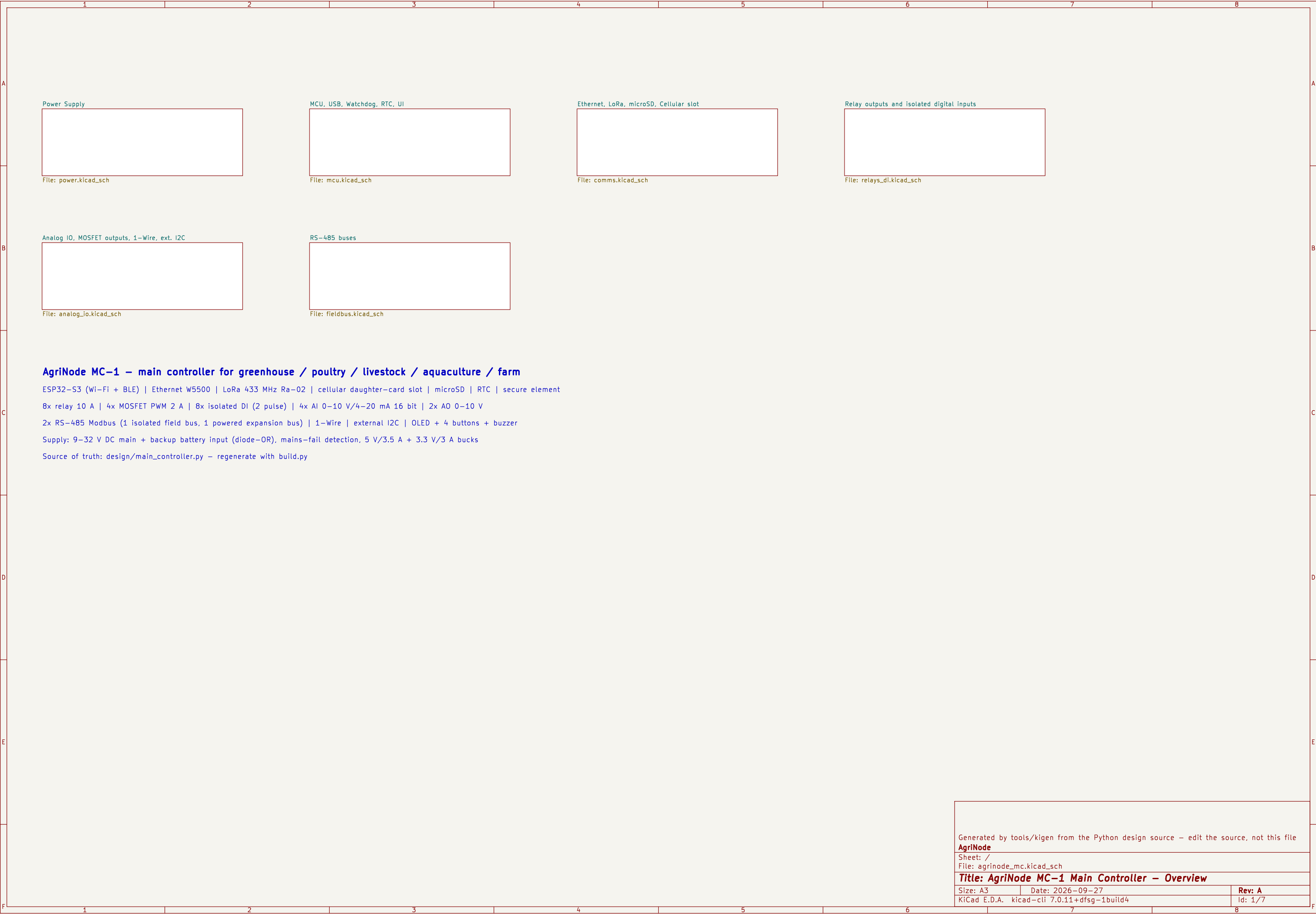
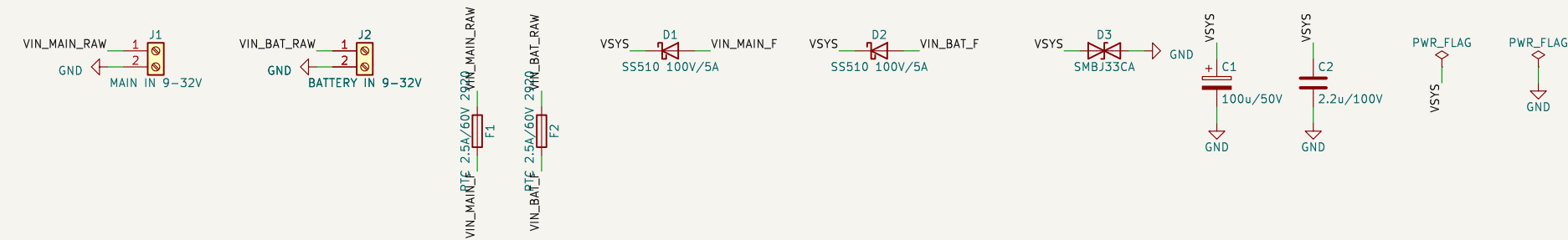


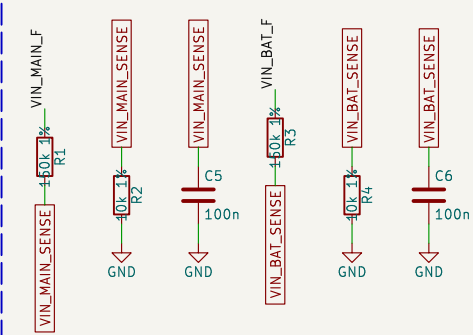


9–32 V DC; each input fused; diode-OR also gives reverse-polarity protection



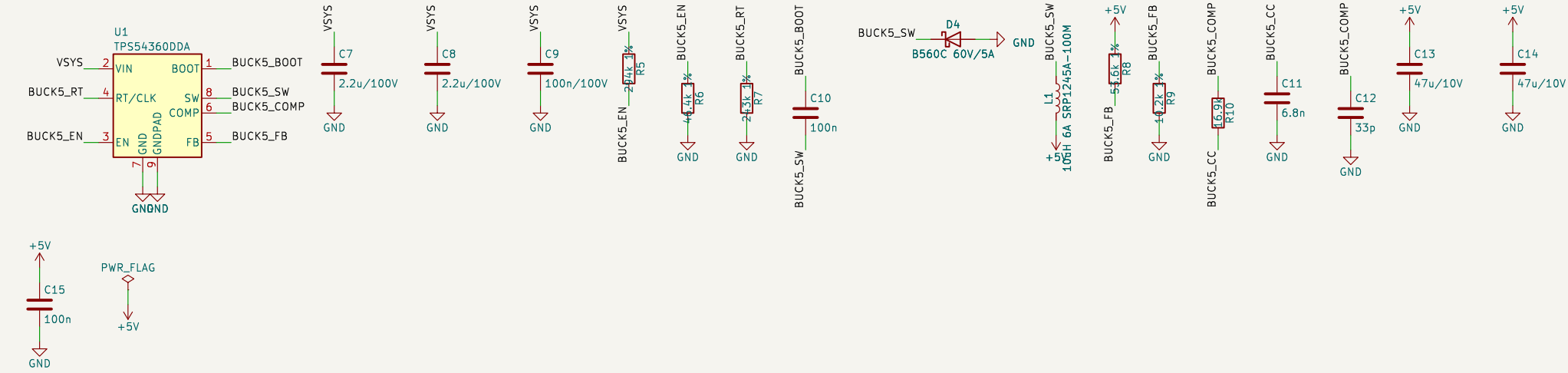
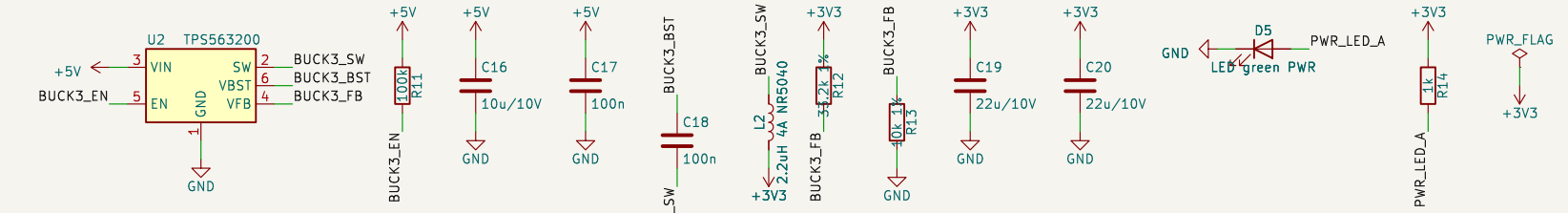
The schematic diagram shows the power supply circuit for the sensor. It includes a 3A/60V 2920 PTC (F3) connected to V5V5, a 0.5A/60V 1812 PTC (F4) connected to V5V5, a 1u/50V capacitor (C3) connected to VFIELD and GND, a 1u/50V capacitor (C4) connected to VSENS and GND, a PWR_FLAG signal connected to VFIELD, and a PWR_FLAG signal connected to VSENS.

150k/10k: 32 V $\rightarrow$ 2.0 V

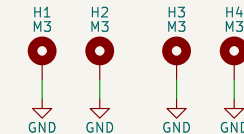
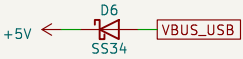


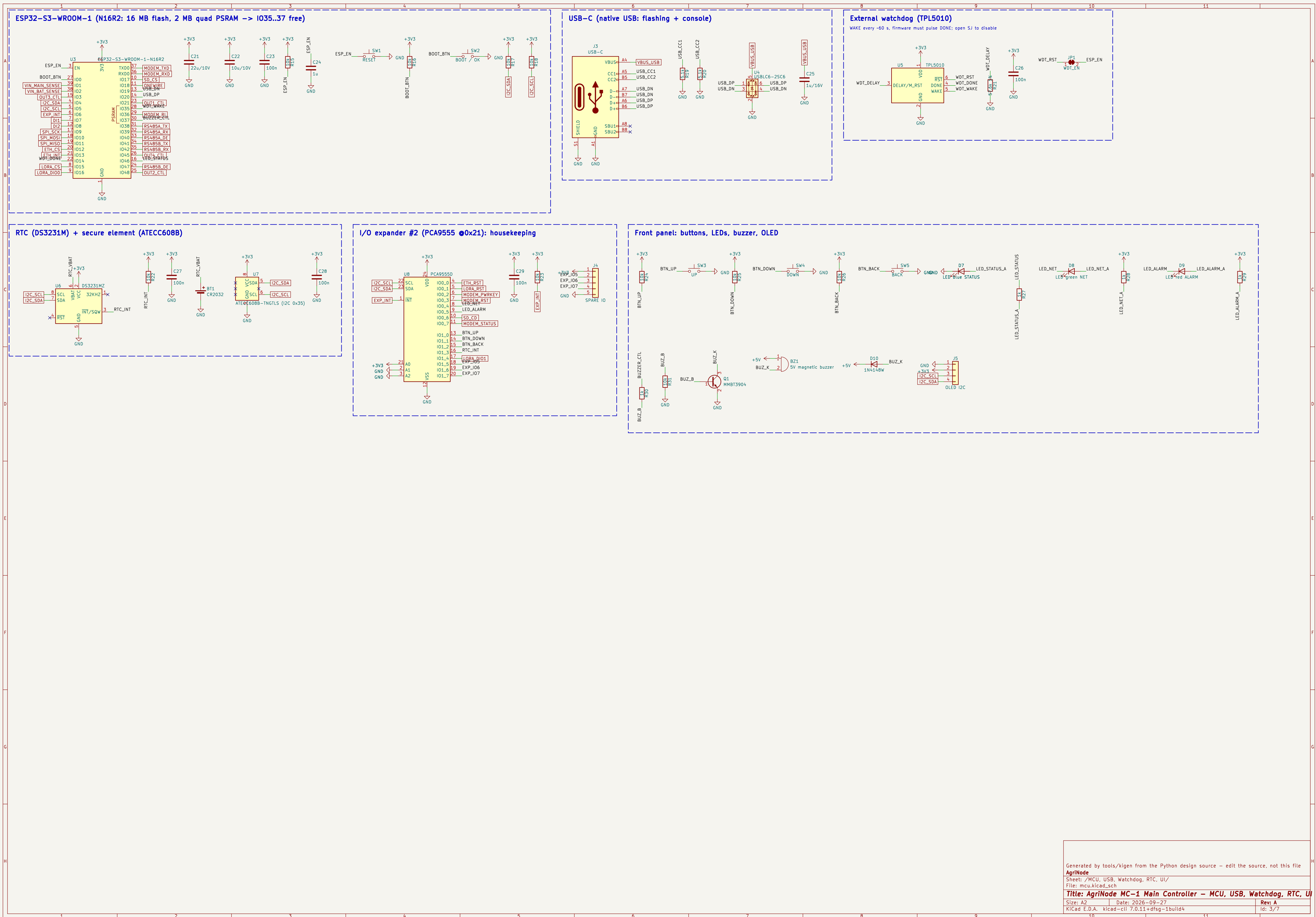
fsw 400 kHz, UVLO on 8.5 V / off 7.5 V

fsw 400 kHz, UVLO on 8.5 V / off 7.5 V

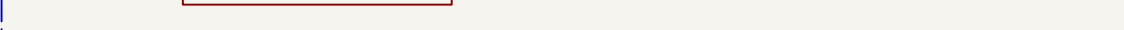
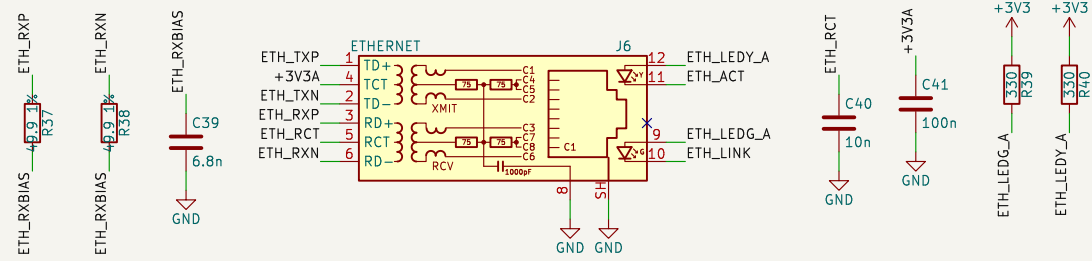

$$V_{out} = 0.768 \times (1 + 33.2k/10k) = 3.32 \text{ V}$$
$$V_{out} = 0.768 \times (1 + 33.2k/10k) = 3.32 \text{ V}$$


USB 5 V can power logic for programming (relays need main supply)



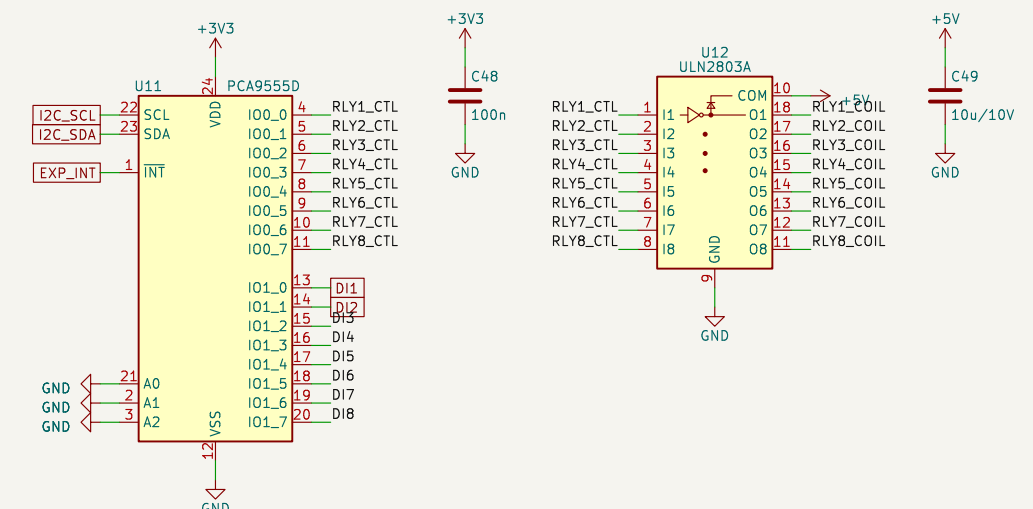


check against WIZnet W5500 reference design

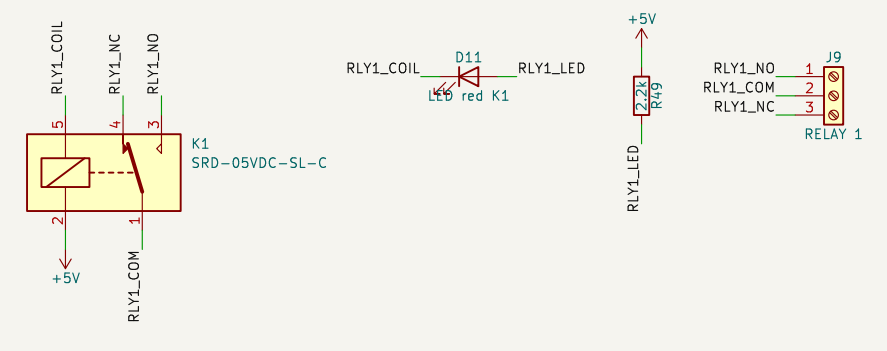


3.3 V logic; card makes its own 3.8–4.2 V from +5V

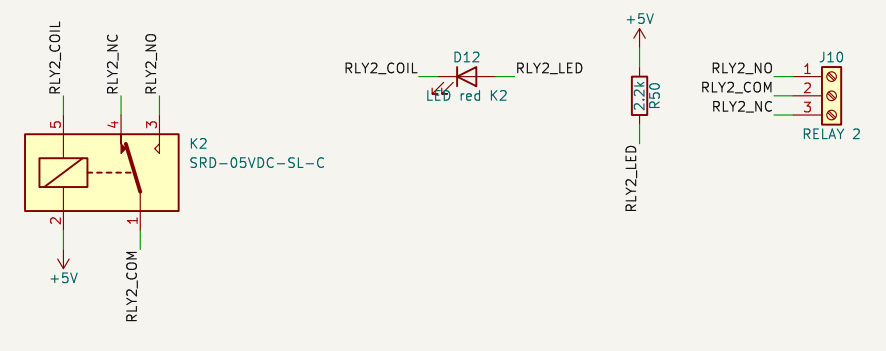
I/O expander #1 (PCA9555 @0x20): relays + digital inputs



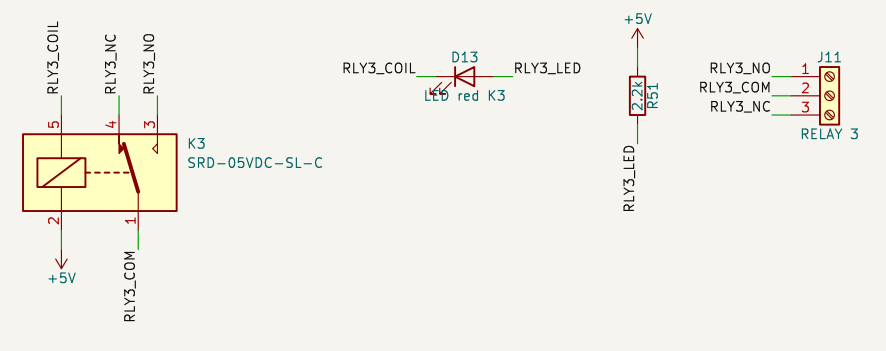
Relay output 1 (10 A 250 VAC, dry contact)



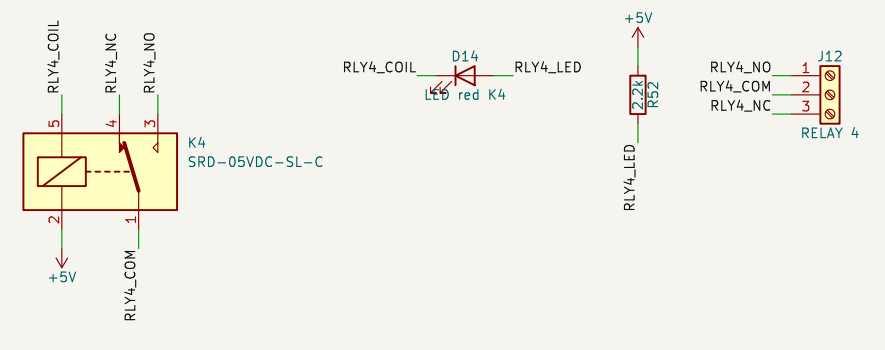
Relay output 2 (10 A 250 VAC, dry contact)



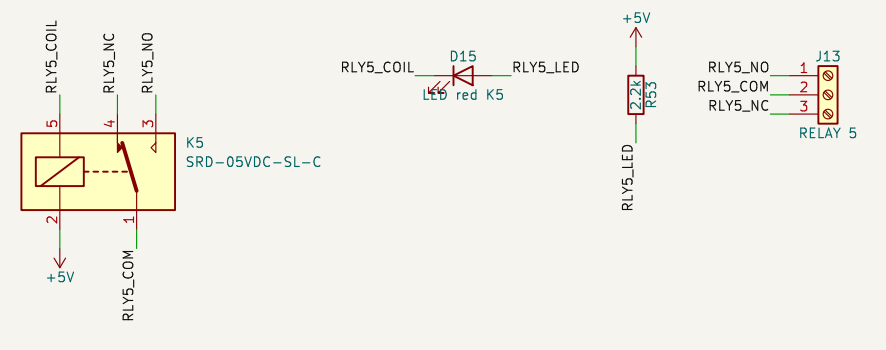
Relay output 3 (10 A 250 VAC, dry contact)



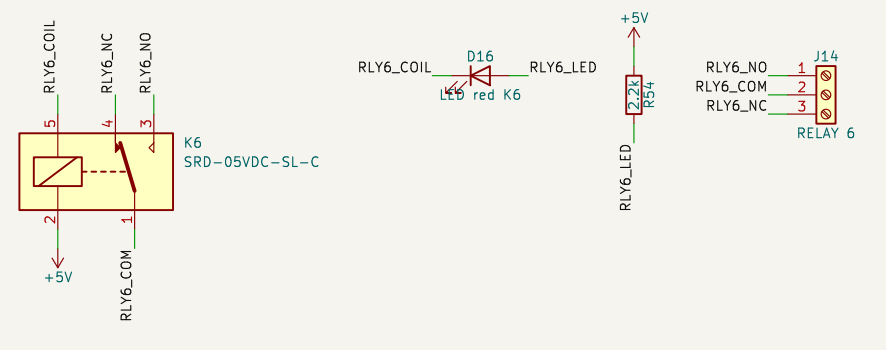
Relay output 4 (10 A 250 VAC, dry contact)



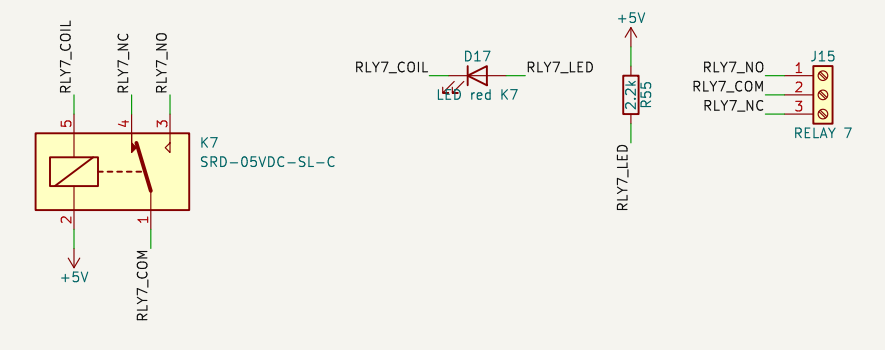
Relay output 5 (10 A 250 VAC, dry contact)



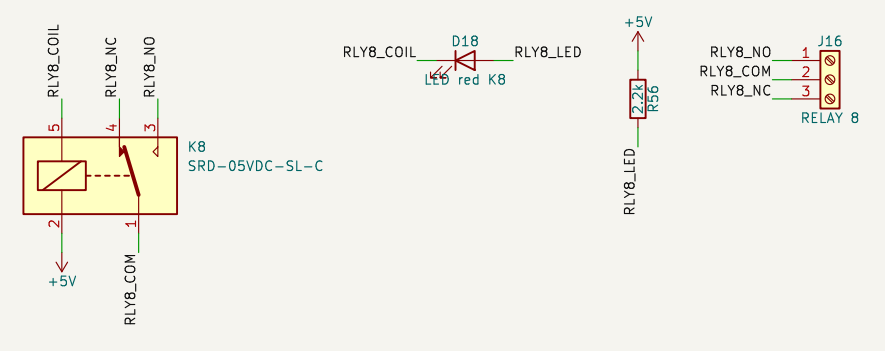
Relay output 6 (10 A 250 VAC, dry contact)



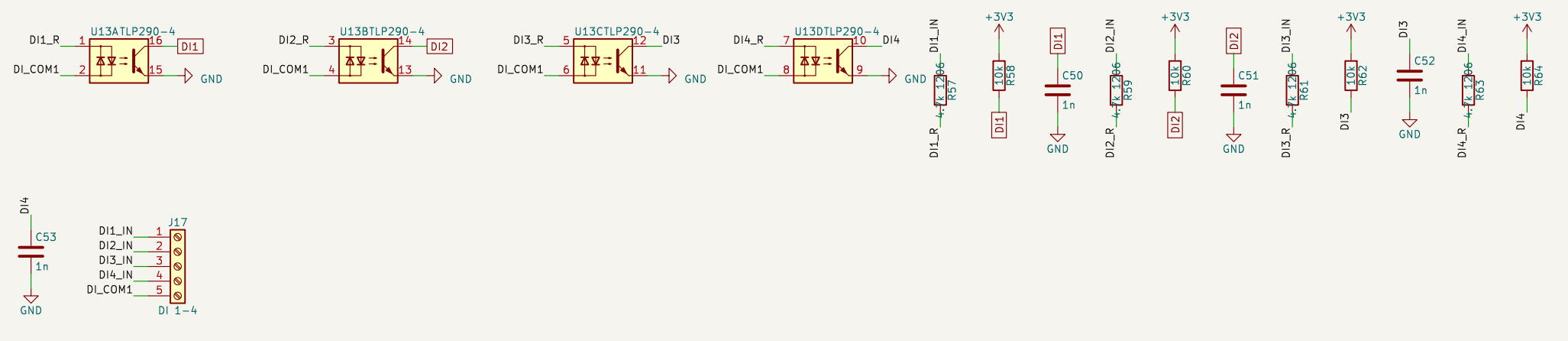
Relay output 7 (10 A 250 VAC, dry contact)



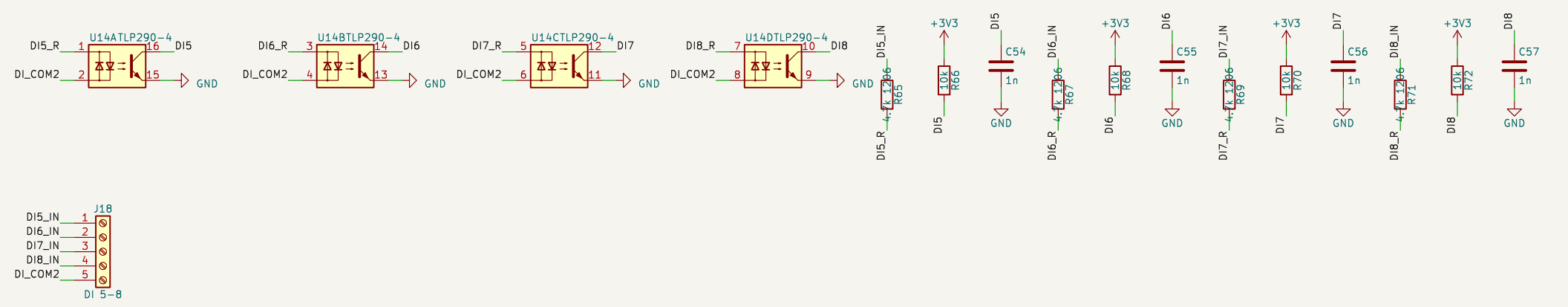
Relay output 8 (10 A 250 VAC, dry contact)

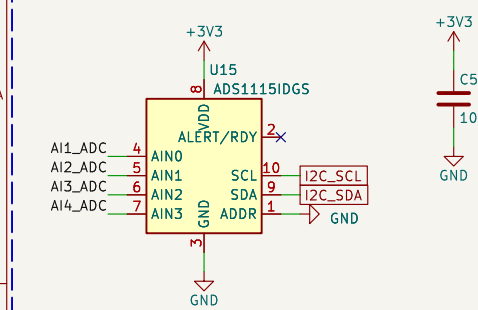
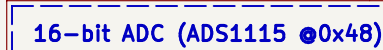


Digital inputs 1–4 (opto-isolated, 10–32 V AC/DC, sink or source)

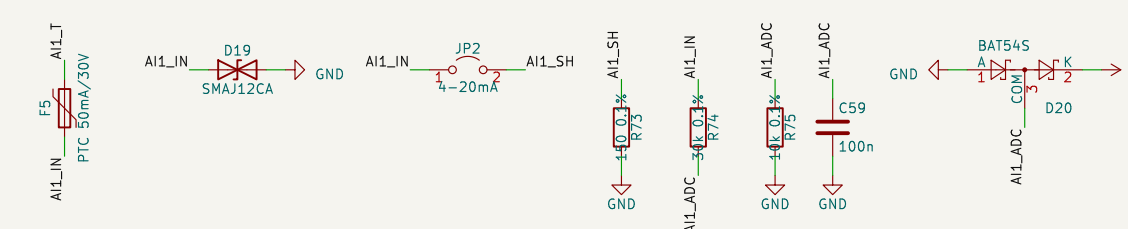


Digital inputs 5–8 (opto-isolated, 10–32 V AC/DC, sink or source)

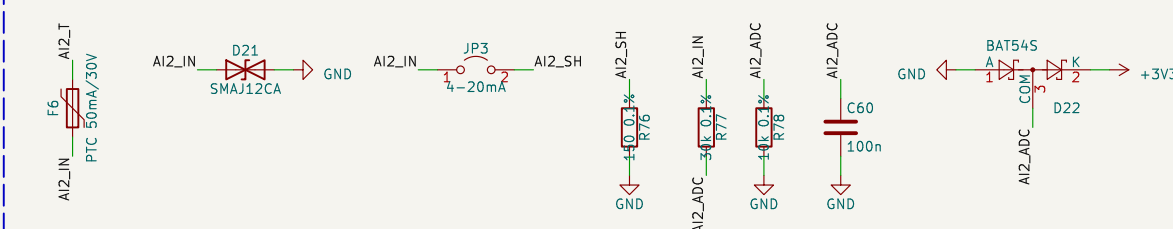




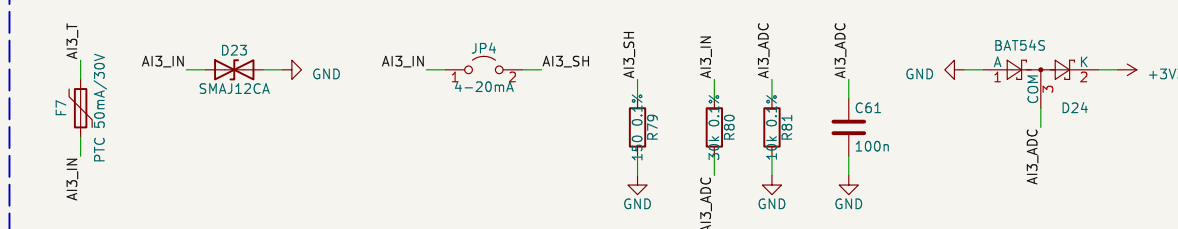
V: 10V $\rightarrow$ 2.5V I: 20mA $\times$ 150R = 3V $\rightarrow$ 0.75V



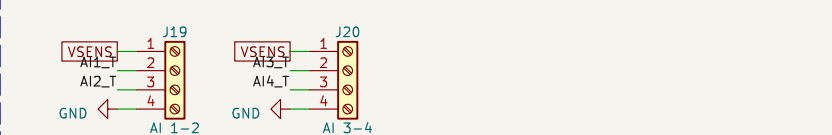
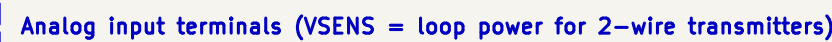
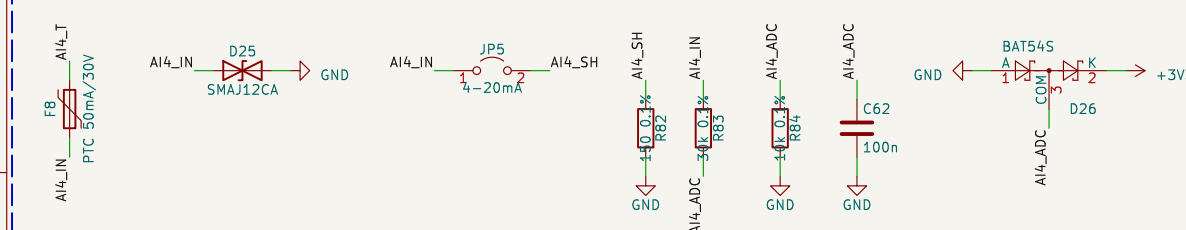
V: 10V $\rightarrow$ 2.5V I: 20mA $\times$ 150R = 3V $\rightarrow$ 0.75V



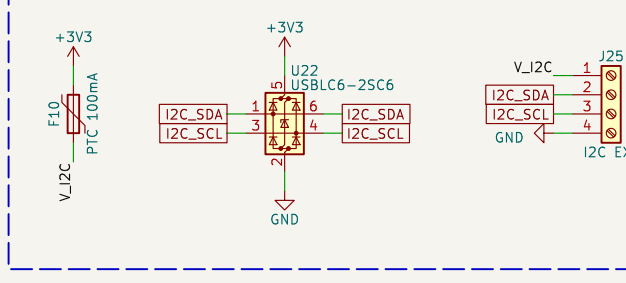
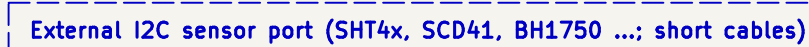
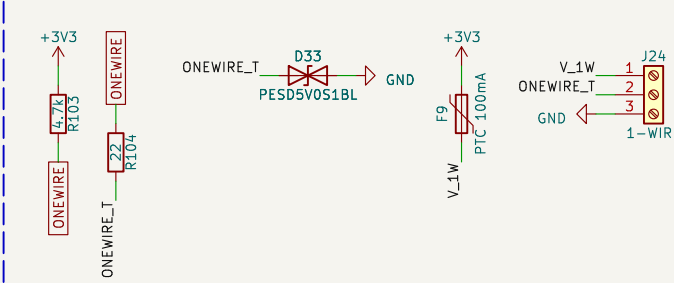
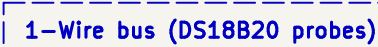
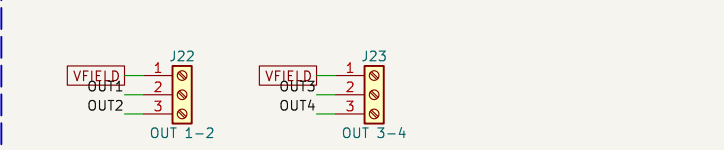
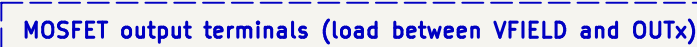
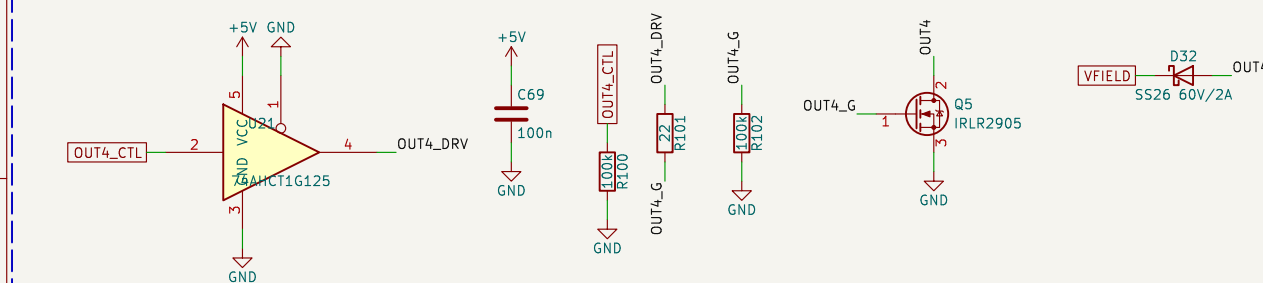
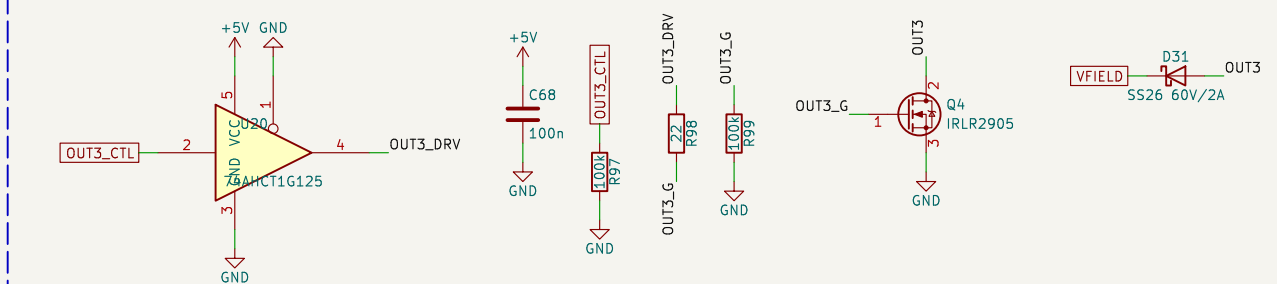
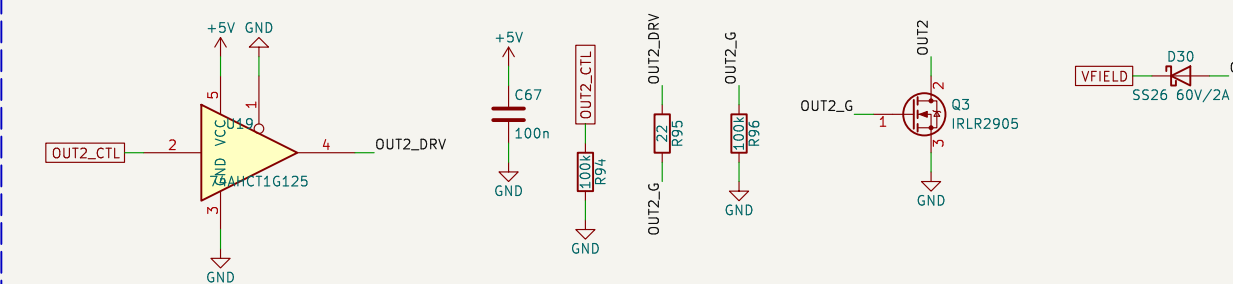
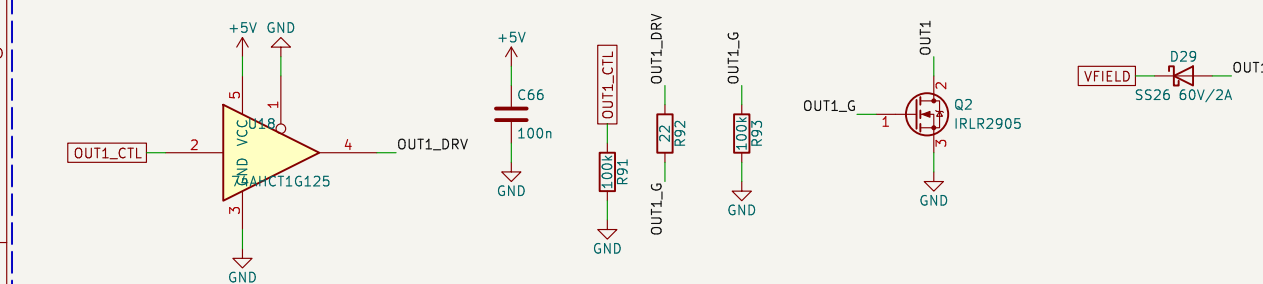
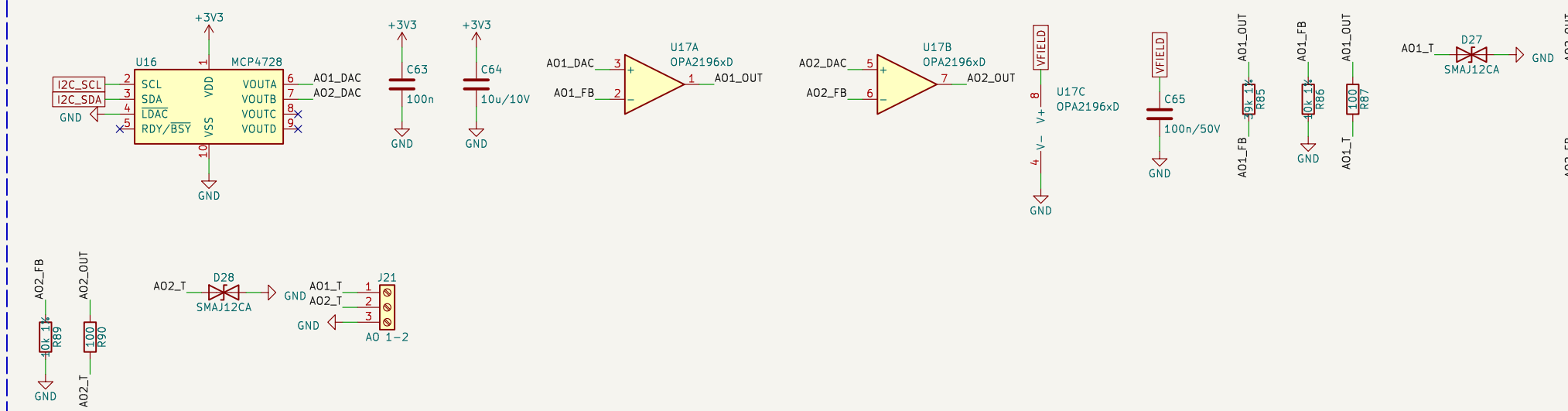
V: 10V $\rightarrow$ 2.5V I: 20mA $\times$ 150R = 3V $\rightarrow$ 0.75V



V: 10V $\rightarrow$ 2.5V I: 20mA $\times$ 150R = 3V $\rightarrow$ 0.75V



needs $V_{SYS} \geq 12\text{ V}$ for full 10 W



RS-485 A: expansion-module bus (Modbus RTU, powered, non-isolated)

The schematic diagram illustrates the internal circuitry of the RS-485 A expansion module. The central component is the THVD1420D driver (U23), which is powered by +3V3 and connected to GND. The driver's VCC pin is connected to +3V3, and its GND pin is connected to GND. The driver's output pins are connected to the RS485_A_TX and RS485_A_RX pins. The RS485_A_TX pin is connected to the driver's A pin, and the RS485_A_RX pin is connected to the driver's B pin. The RS485_A_DE pin is connected to the driver's DE pin. The RS485_A_TERM pin is connected to the driver's TERM pin. The module is powered by +3V3 and GND. The RS485_A_TX and RS485_A_RX pins are connected to the RS485_A_EXP BUS via J26. The RS485_A_TERM pin is connected to the RS485_A_TERM connection via J6. The RS485_A_DE pin is connected to the RS485_A_DE connection via J6. The RS485_A_TERM pin is connected to the RS485_A_TERM connection via J6. The RS485_A_TX and RS485_A_RX pins are connected to the RS485_A_EXP BUS via J26. The RS485_A_DE pin is connected to the RS485_A_DE connection via J6. The RS485_A_TERM pin is connected to the RS485_A_TERM connection via J6.

